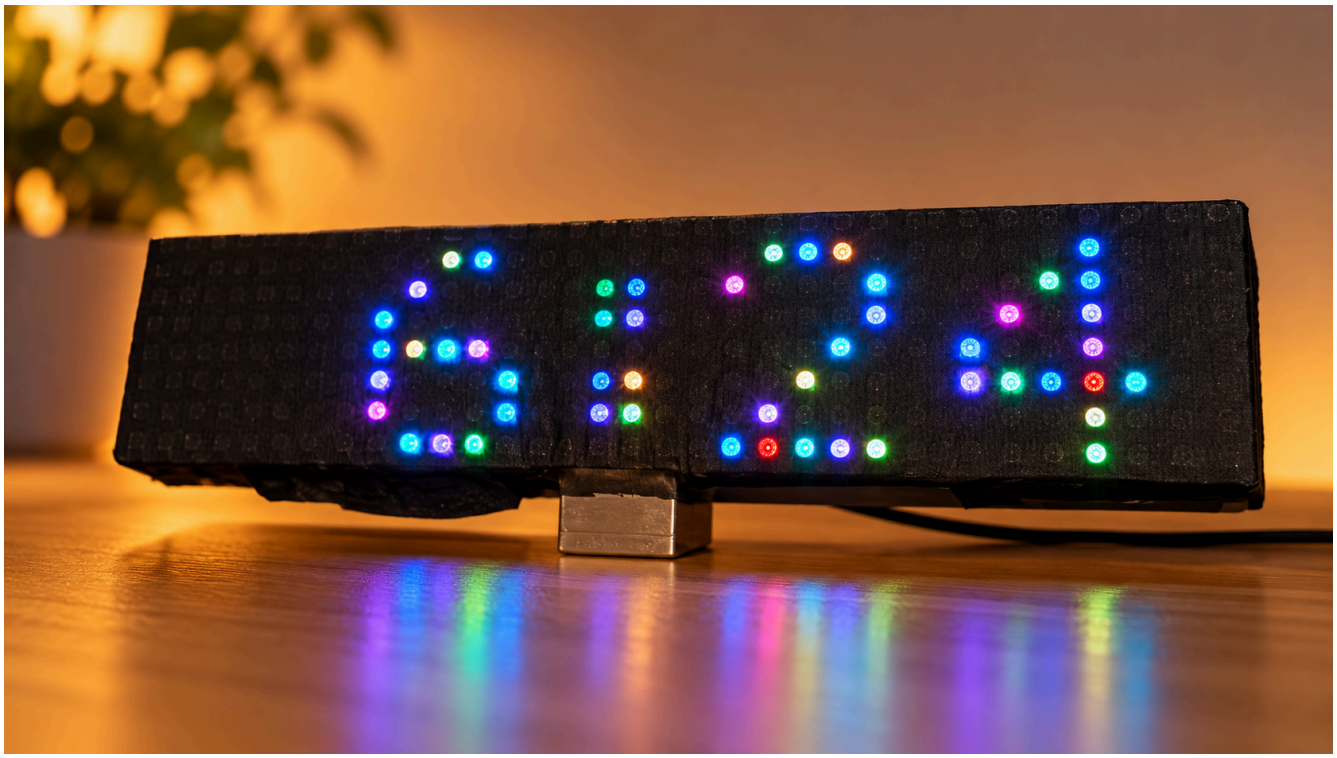
ESP Siesta Clock with Matrix Display

User Manual - English



A clever, super-dark clock that respects your sleep.

1. What this clock is

The ESP Siesta Clock is a WiFi-assisted RGB matrix clock built for comfortable everyday use, especially at night. The 32 x 8 WS2812 matrix shows the time; a small OLED is used only for menus, status and setup. A DS3231 keeps time continuously, so the clock does not depend on a permanent Internet connection.

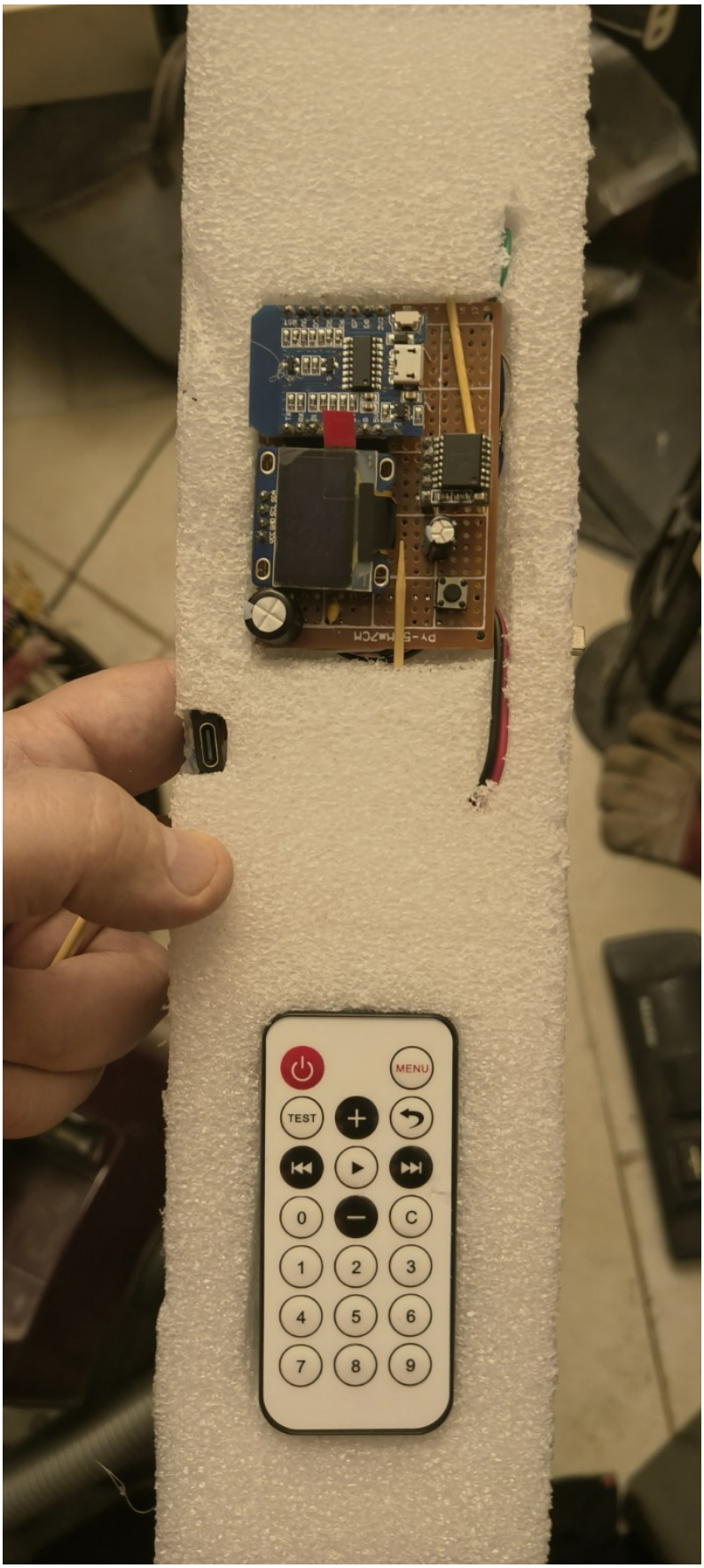
Its defining feature is its sleep-first behavior: during Night Mode the display brightness is forced to the minimum setting and other display effects are not allowed to make it brighter.

2. Everyday use

- Power the clock normally. It reads its saved settings and DS3231 time.
- If saved WiFi is available, the clock makes a limited connection attempt, synchronizes time, updates the RTC, refreshes public-holiday data when required, and then turns WiFi off.
- If WiFi is not available, the clock continues from the DS3231 without stopping normal time display.
- The OLED normally remains off outside setup/status activity; this is normal.

3. Remote control

Key / label	Action
MENU	Open the SIESTA CLOCK editor
+ / UP	Move up or increase a value
- / DOWN	Move down or decrease a value
Previous / <<	Move cursor left
Next / >>	Move cursor right
Play / ENTER	Open, toggle, confirm or save
Return arrow / BACK	Go back
C / CANCEL	Cancel / leave an edit
0-9	Enter time/date digits directly
TEST / Power	Show clock information/status



The electronics and the NEC remote used by the photographed build.

4. Main menu

Menu item	What it changes
Country	Country used for public-holiday data
Time Zone	Civil-time / DST rule
Time	Manual RTC time
12/24 Hour	Clock display format
Fonts	One of six matrix font styles
Siesta	Summer/Winter quiet-hour schedules
Night Mode	Normal dim start/end
Holidays	Public/Saturday/Sunday override rules
Dimmer	Day brightness, 1-30
AUTOSYNC	Automatic synchronization interval, 1-255 hours
SYNC NOW	Immediate WiFi/NTP/RTC synchronization
Mono Color	Select one of 16 fixed colors
Multi Color	Solid, Colorful, Funny or Dynamic
Language	English or Greek OLED UI

5. Night Mode - the super-dark mode

The default Night Mode is 23:00 to 07:00. During this period the firmware forces brightness to 1. This rule has priority: menu color overrides and normal daytime effects cannot raise the brightness while Night Mode is active. Day brightness is adjustable from 1 to 30; the fresh-install default is 6.

6. Holidays and weekends

Turn Holidays On to open four options: Public Holidays, Saturdays, Sundays and Time. The three date rules are independent checkboxes. If any selected rule matches today, the configured Holidays time replaces only the normal Night Mode end time. The Night Mode start time and all other clock behavior remain unchanged.

Example: Night Mode 23:00-07:00, Holidays time 08:30, Sundays enabled. On Sunday the clock remains at its minimum night brightness until 08:30.

7. Siesta / quiet hours

Season	Day quiet period	Night quiet period	Dates
Summer	15:00-17:30	23:00-07:00	01/04-30/09
Winter	15:30-17:30	22:00-07:30	01/10-31/03

The Siesta editor uses eight screens. The complete Summer/Winter configuration is validated before it is stored, so an invalid unfinished schedule does not replace the saved configuration.

8. WiFi setup

- To configure or replace WiFi credentials, hold the hardware configuration button during boot.
- The OLED shows the access point name AutoConnectAP.
- Connect a phone/computer to AutoConnectAP and open 192.168.4.1.
- Enter the WiFi credentials. On normal boots the clock does not enter the configuration portal automatically.

9. Time synchronization

AUTOSYNC is adjustable from 1 to 255 hours. The internal firmware default is 24 hours. SYNC NOW starts an immediate connection and synchronization. After synchronization the WiFi radio is shut down again. If synchronization fails, the DS3231 remains the time source.

10. Display styles and colors

- 6 selectable clock font styles.
- 12-hour or 24-hour time display.
- 16 mono colors.
- Multi Color modes: Solid Color, Colorful, Funny and Dynamic.
- A breathing two-column separator sits between hours and minutes.
- In 12-hour mode, a PM indicator pixel is used for PM times.

11. Troubleshooting

Symptom	What to check
Clock works but does not sync	Saved WiFi, router/Internet, then try SYNC NOW
Wrong local time	Check Time Zone; Country and Time Zone are intentionally separate
Public holiday has no effect	Holidays On, Public checked, correct Country, holiday download/cache
Weekend has no effect	Enable Saturday and/or Sunday in Holidays
Very dim at night	Expected: Night Mode forces brightness 1
No WiFi at startup	The clock should continue from DS3231; this is intentional
Need new WiFi credentials	Hold the hardware config button during boot and use AutoConnectAP

12. Firmware identity

This manual describes ESP_Siesta_Clock_with_Matrix_Display.ino. Publication source: 7,979 lines, 177,326 bytes, SHA-256 8e842bcc9da1b2affbb206ff4803113f7d8c407f40a1ed76053045c1f0a75b74.